

Prompt Gamma Spectroscopy for Real-Time Calcium Quantification in Proton Therapy

Mats Stauske^{1,2}, Hugo Freitas¹, Mariana Brás^{1,3,4}, Ömer Nuhoğlu^{1,2}, João Seco^{1,2}

¹ Division of Biomedical Physics in Radiation Oncology, DKFZ Heidelberg, Germany, ² Department of Physics and Astronomy, Heidelberg University, Heidelberg, Germany, ³ Laboratory of Instrumentation and Experimental Particle Physics (LIP), Lisbon, Portugal, ⁴ Department of Physics, Instituto Superior Técnico, University of Lisbon, Lisbon Portugal

Background

- Bone radiotherapy can induce **localized changes in mineral content** in irradiated bone, evolving **during treatment** and between fractions. Monitoring these changes remains challenging with standard imaging.^[1]
- Prompt Gamma Spectroscopy (PGS)** could provide **real-time calcium (Ca) mass-density quantification** as a direct metric of treatment response.
- Prompt gammas (PG)**: Proton-induced inelastic nuclear scattering populates excited nuclear states, which de-excite via gammas on ps–ns timescale.
- Clinical context**: Prompt gammas can potentially be used for in vivo range verification to reduce range uncertainty and treatment margins. Element-specific prompt-gamma lines may additionally enable elemental analysis.^[2] **Key benefit**: Elemental information without additional patient dose.

Materials and Methods

- Monte Carlo simulations in **TOPAS** with a 100 MeV transmission proton pencil beam and 10^{10} protons irradiating **two series of cylindrical phantoms**.^[3,4]
- A lateral **CeBr₃** scintillator recorded the prompt-gamma spectrum, which was convoluted with an energy-dependent resolution function and ideal anticoincidence (AC) was applied to suppress escape peaks.
- Phantoms: Calibration + validation phantoms
 - Calibration: 0–100 wt% Ca;
 - Validation: CT phantoms with tissue-equivalent compositions.

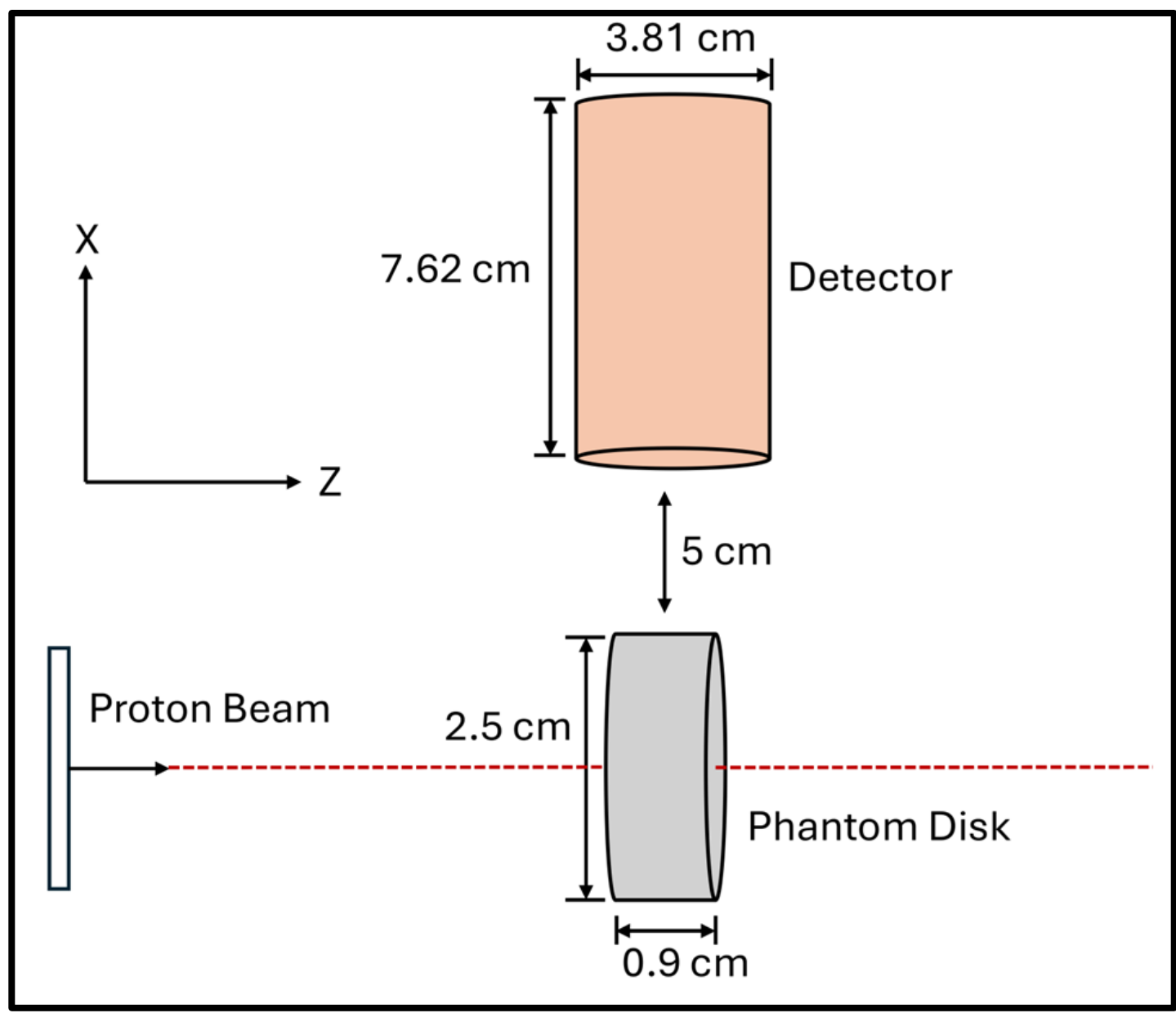


Figure 1: Schematic of the simulation geometry used for the two sets of simulations.

The Multi-peak Fit Model

- Areas from **six Ca prompt-gamma lines** were extracted via **background-subtracted** integration over predefined energy windows.
- Each peak showed an approximately **linear** dependence on Ca content ($R^2 > 0.996$), enabling linear-regression-based Ca estimates.
- Individual peak estimates were combined via **weighted fusion** into a single Ca prediction to improve robustness over any single line.

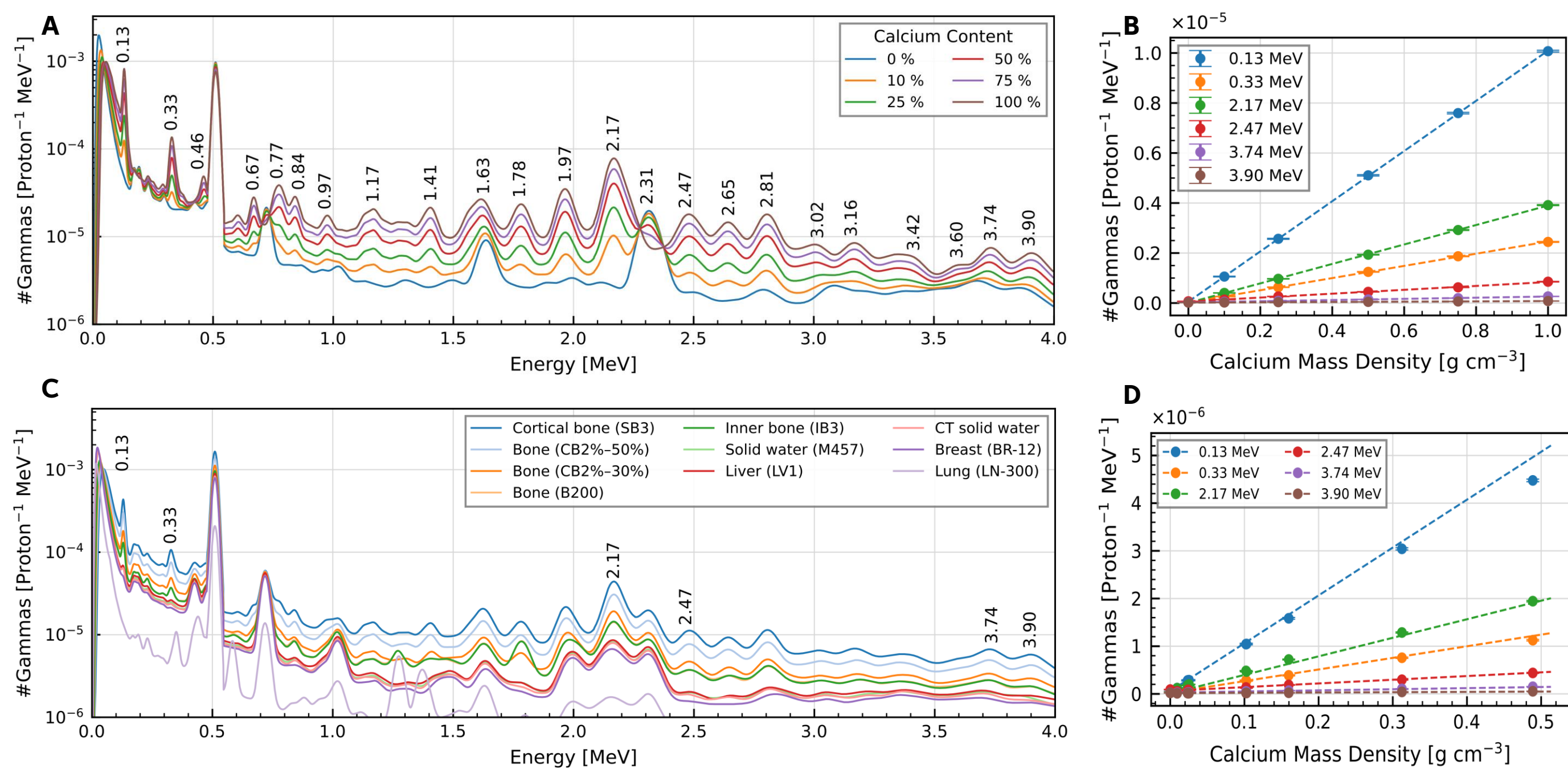


Figure 3: Simulated gamma spectra for Ca-calibration-mixtures and tissue-equivalent validation materials, with linear peak–Ca relationships for six Ca lines. (A) Calibration spectra. (B) Peak area vs Ca (linear fits). (C) Tissue-equivalent spectra. (D) Overlay of tissue-equivalent peak areas on calibration fits.

Identified Nuclear Reaction Channels

Nuclear reaction identification via Geant4 particle tracking cross checked with NuDat3 database and TALYS calculations.^[5–9] Multiple nuclear reaction pathways contribute to characteristic spectral peaks.

Peak selection criteria for calcium quantification:

- Minimal overlap with O, C, N, and P emission lines;
- Strong emission yield and high prominence above background.

Six calcium-specific lines identified satisfying the criteria: 0.13, 0.33, 2.17, 2.47, 3.74, and 3.90 MeV.

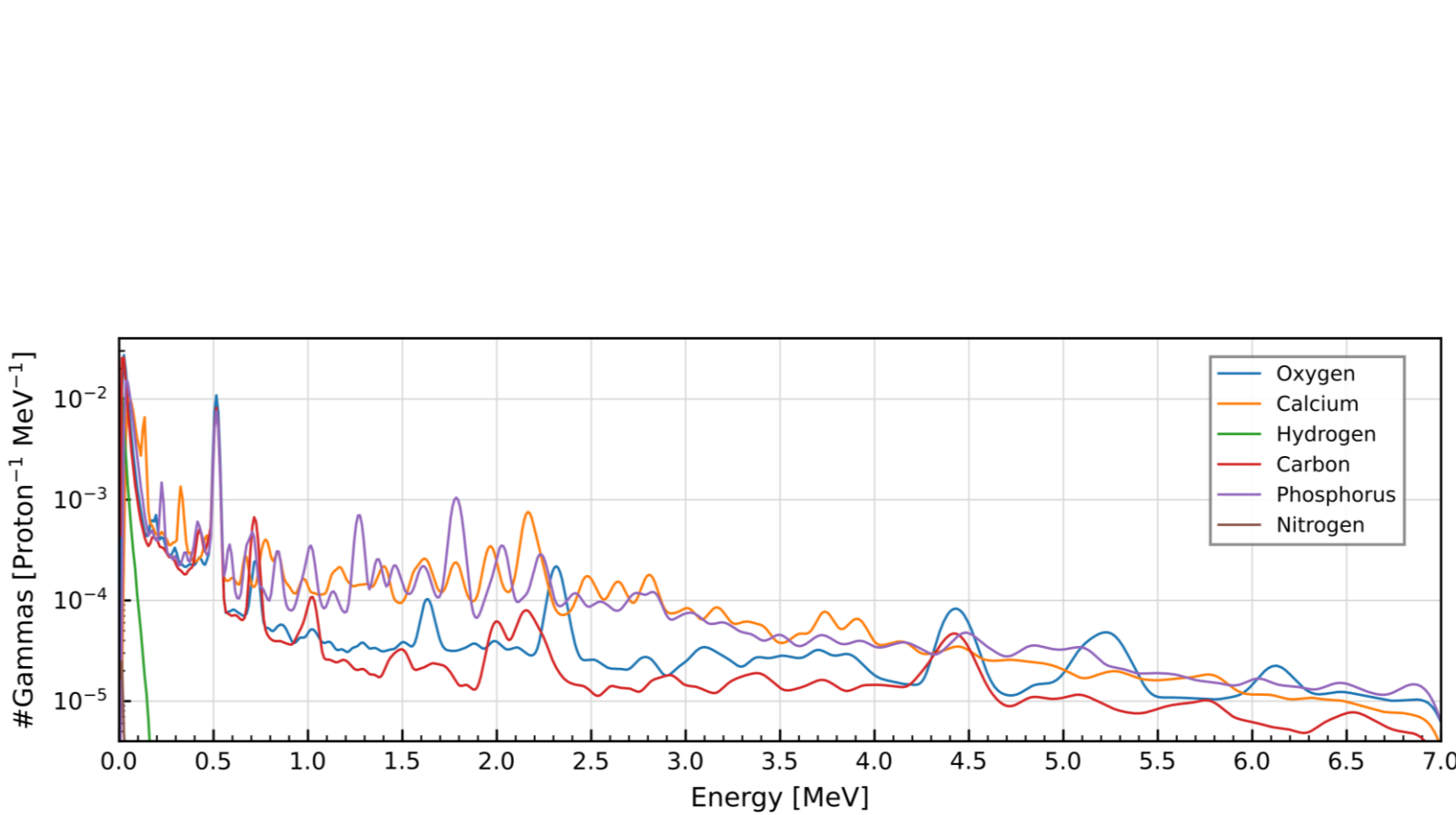


Figure 2: Simulated gamma spectra for pure elements (O, C, H, N, P, and Ca) with identical densities under 100 MeV proton irradiation.

E_{peak} [MeV]	E_{γ} [MeV]	Nuclear Reaction
0.13	0.130	$^{38}\text{K}_{0.130}^* \rightarrow ^{38}\text{K}_{g.s.}$
0.33	0.328	$^{38}\text{K}_{0.459}^* \rightarrow ^{38}\text{K}_{0.130}$
	2.167	$^{38}\text{Ar}_{1.410}^* \rightarrow ^{38}\text{Ar}_{g.s.}$
2.17	2.127	$^{34}\text{S}_{2.127}^* \rightarrow ^{34}\text{S}_{g.s.}$
	2.208	$^{36}\text{Ar}_{4.178}^* \rightarrow ^{36}\text{Ar}_{1.970}^*$
	2.217	$^{37}\text{Ar}_{2.217}^* \rightarrow ^{37}\text{Ar}_{g.s.}$
2.47	2.467	$^{39}\text{Ca}_{2.467}^* \rightarrow ^{39}\text{Ca}_{g.s.}$
	2.523	$^{39}\text{K}_{2.523}^* \rightarrow ^{39}\text{K}_{g.s.}$
3.74	3.737	$^{40}\text{Ca}_{3.737}^* \rightarrow ^{40}\text{Ca}_{g.s.}$
3.90	3.904	$^{40}\text{Ca}_{3.904}^* \rightarrow ^{40}\text{Ca}_{g.s.}$

Table 1: Prompt-gamma transitions from proton-induced reactions on ^{40}Ca .^[8]

Results and Conclusion

- Prediction performance**: sub-1 wt% Ca prediction precision using the 6-line multi-peak + fusion approach on validation data set spectra.
- Multi-peak fusion outperforms single line approaches and improves robustness** to detector-resolution broadening and peak overlap with neighboring non-Ca peaks.
- Conclusion**: This supports the technical feasibility of real-time Ca quantification with PGS, with potential future clinical use for lesion-specific Ca tracking and adaptive strategies in proton therapy of bone cancer and bone metastases.

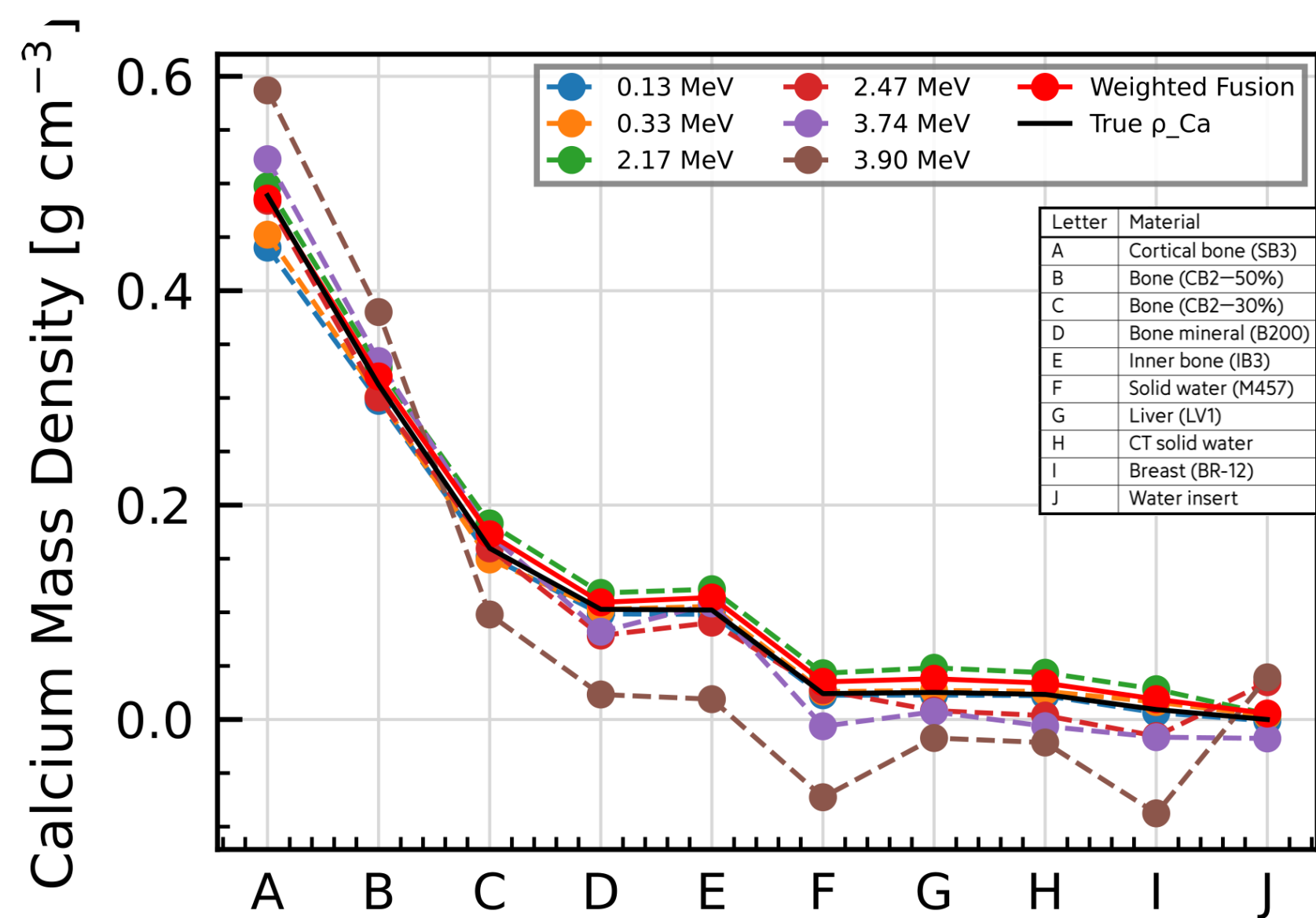


Figure 4: Predicted vs reference Ca mass density for tissue-equivalent materials.

References,
Declaration
and more
Information:



Future Work & Outlook

- Simulation refinements**: Include a more realistic beam energy modulation, spectral attenuation and energy straggling via tissue-equivalent shielding materials, surrounding the samples and realistic AC.
- Experimental validation**: Measurements using calcium-containing materials (e.g., hydroxyapatite, β -TCP) and tissue-equivalent Gammex inserts to confirm simulated spectral features.